Edgar Perez Vidal

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Education

Tufts University Medford, MA
Ph.D. Astrophysics Expected 2029
M.S. Astrophysics 2026

University of California, Berkeley Berkeley, CA
B.A. Physics; B.A. Astrophysics 2023

Research Appointments

Graduate Researcher, Tufts University Medford, Ma
Advisors: Anna Sajina and Danilo Marchesini 2024-present
Census of obscured Active Galactic Nuclei with *JWST*.

Undergraduate Researcher, UC Berkeley Berkeley, CA
Advisor: Prof. Alex Filippenko; Co-advisor: Kishore Patra 2022–2024
Observational Transient-Based Astronomy and Nickel 1-meter telescope operator

Research Intern, IRAP Toulouse, France
Advisors: Florian Sarron & Nicolas Clerc June–Aug 2023
Galaxy cluster mass and cosmic web connectivity with SDSS and XCLASS.

Publications

First Author

1. **Vidal, E. P.**; Sajina, A.; Banks, A. R.; Bethermin, M.; Ferkinhoff, C.; Petric, A.; Pope, A.; Lyu, J.; U, V.; Yung, L. Y. A.; Patil, P. (2026). **Modeling the JWST MIRI Counts, Insights Into the Source Properties and Role of Dust-Obscured AGNs**. *ApJ*, **1002**, 23.

N-th Author

7. Cutler, S. E. et al. (including **Vidal, E. P.**) (2026). **Unbreak the Universe: MINERVA Color Gradients in Massive Quiescent Galaxies Can Resolve Too-Early Star Formation Tensions**. *arXiv*, arXiv:2606.02698.

6. Hamblin, K. et al. (including **Vidal, E. P.**) (2026). **AGNBoost: A Machine Learning Approach to AGN Identification with JWST/NIRCam+ MIRI Colors and Photometry**. *ApJ*, **1002**, 223.

5. Gendreau-Distler, E. A. et al. (including **Vidal, E. P.**) (2025). **Transit Timing of the White Dwarf–Cold Jupiter System WD 1856+534**. *arXiv*, arXiv:2511.21611.

4. Muzzin, A. et al. (including **Vidal, E. P.**) (2025). **MINERVA: A NIRCam Medium Band and MIRI Imaging Survey to Unlock the Hidden Gems of the Distant Universe**. *arXiv*, arXiv:2507.19706.

3. Zheng, W. et al. (including **Vidal, E. P.**) (2025). **SN 2023ixf in the Pinwheel Galaxy M101: From Shock Breakout to the Nebular Phase**. *ApJ*, **988**, 61.

2. Alvarado, E. et al. (including **Vidal, E. P.**) (2024). **Searching for Tidal Orbital Decay in Hot Jupiters**. *MNRAS*, **534**, 800–813.

1. Kong, D.-F. et al. (including **Vidal, E. P.**) (2024). **GRB 221009A/SN 2022xiw: A Supernova Obscured by a Gamma-Ray Burst Afterglow?** *ApJ*, **971**, 56.

Workshop & Conference Proceedings

1. **Vidal, E. P.**; Gagliano, A. T.; Cuesta-Lazaro, C. (2025). **Hierarchical Simulation-Based Inference of Supernova Power Sources and Their Physical Properties**. *Machine Learning for the Physical Sciences Workshop at NeurIPS 2025*. arXiv:2510.14202.

Invited Talks

University of Kansas Astro/Space Seminar	2026
Harvard & Smithsonian CfA Transient Tea	2025
Harvard & Smithsonian CfA AstroAI Lunch Talk	2025
Harvard & Smithsonian CfA UMBRELA Dialogue	2025
IPAC PRIMA P-CAST Talk	2025

Teaching Experience

Graduate Teaching Assistant, Tufts University — 2024–2025

- Introductory Physics 1/11 & Physics 2/12 Laboratory
- Introductory Physics 2 & 11 Recitation

Undergraduate Graduate Student Instructor, UC Berkeley — Aug–Dec 2023

Introduction to General Astronomy, Astro C10, *under Prof. Alex Filippenko*

Course Reader, UC Berkeley — 2022, 2024

- Stellar Astrophysics, Astro 160, *under Prof. Raffaella Margutti*
- Relativistic Astrophysics and Cosmology, Astro C161, *under Prof. Daniel Kasen*

Mentorship

Undergraduate Students

Amber R. Banks (Tufts) — 2025

- Building an AGN SED library and color-color diagrams for selecting AGN
- 3rd author on *ApJ* **1002**, 23; presented results at Tufts Astro Seminar.

Service & Outreach

Inclusive Excellence in Physics and Astronomy Committee Member — 2025–Present

Tufts University, Department of Physics & Astronomy

- **Graduate–Undergraduate Mentorship Program** (Founder & Lead)
  Launched the department's first peer mentorship network, connecting 50+ students across research, graduate school preparation, and academic life.

Astronomy Graduate Student Congress — 2024–Present

National

- Promoting interdepartmental discourse among graduate students nationwide; developing outreach initiatives and centralizing grad school resources for early-career scientists.

Honors & Awards

- **First Place**, Astromatic Hackathon, Ciela Institute, Université de Montréal — 2023
  Simulation-Based Inference Model for Gravitational Lensing Parameters Estimation
- LSAMP NSF International Center of Excellence (NICE) Fellowship — 2023
- NSF CAMP/LSAMP Research and Travel Fellowship — 2022, 2023
- NSF CAMP Statewide Symposium Best Presentation Award — 2023

- SACNAS NDISTEM Travel Grant 2022, 2023
- Rose Hill Foundation Scholarship 2019
- California Middle-Class Scholarship 2019
- Berkeley CARES Award 2019
- Berkeley Scholarship 2019
- Simon Family Foundation Scholarship 2018

Conferences & Posters

Contributed

American Astronomical Society Meeting #247 2026

NeurIPS, Machine Learning for the Physical Sciences Workshop 2025

NSF CAMP Statewide Symposium 2024

NSF CAMP Statewide Symposium (**Special Merit Award**) 2023

SACNAS NDISTEM Conference 2022, 2023

Professional Experience

Student Director, Cal NERDS 2023–2024

- Co-instructed a weekend Python bootcamp for 50+ STEM students; led the Research Poster Design and Printing Lab Project.

Academic Mentor, Calculus Round Table 2021–2022

- Tutored K–12 students in biology, math, astronomy, and Python; developed remote curriculum during COVID-19.
- Mentored students at the Juvenile Justice Center in San Leandro, CA.

Kitchen Manager, Berkeley Student Cooperative 2021–2022

- Managed a $20,000 food and supplies budget for 30 members. *ServSafe Certified.*

R.I.S.E Mentor, Berkeley High School 2020–2021

- Mentored students from underprivileged backgrounds toward college readiness.